Check for updates

Original Article

POMS

Production and Operations Management
2026, Vol. 35(10) 3715–3729
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10591478261442318
journals.sagepub.com/home/pao

Sage

Why full refunds prevail: A product fit perspective

Weitao Ren[1], Chuanshuai Ru[1], Wenqiang Xiao[2] and Fangruo Chen[1]

Abstract

This paper studies how an online seller designs a menu of return contracts to manage consumer heterogeneity arising from product misfit risk. In the model, informed consumers know the product fits, whereas uninformed consumers face uncertainty about fit. We show that when only misfit risk exists, a single full-refund contract can implement the optimal menu. The return price insures consumers against misfit, and the selling price extracts full surplus. This finding aligns with the widespread adoption of lenient return policies and demonstrates that uniform contracts can emerge endogenously from incentive compatibility rather than as ad hoc assumptions. Introducing quality risk—an additional source of uncertainty in the valuation of fit products—fundamentally changes this outcome. The seller then shifts from full to partial refunds. Under high quality risk, the optimal menu becomes differentiated: A low-price, no-frills option for informed consumers and a high-price, insurance-heavy option for uninformed consumers. To induce self-selection, the seller distorts the return price upward for uninformed consumers to strengthen the insurance effect, trading off allocative efficiency for screening. Interestingly, when quality risk becomes extreme, refunds not only mitigate information rents but also enhance social surplus by preventing consumers from retaining low-quality products. Extensions incorporating misfit valuation, seller-side costs, and return hassle costs show that the uniform contract remains robust when only misfit risk is present, rationalizing diverse return policy practices in online markets.

Keywords

Online Market, Full Refunds, Return Policy, Consumer Heterogeneity, Product Misfit

Received: 21 April 2025; accepted: 26 March 2026 after two revisions

Handling Editor: Albert Ha

1 Introduction

Today's online retail market generally consists of two distinct consumer types: Loyal, experienced buyers and inexperienced newcomers, hereafter referred to as informed and uninformed consumers, respectively. Industry evidence illustrates this segmentation clearly. In fashion retail, for example, informed consumers represent 54% of online buyers and repurchase familiar products with confidence, while the remaining 46% lack such product-specific knowledge and often face uncertainty about product fit (Gardner, 2025). A garment that looks appealing online may disappoint only after it is tried on, exposing uninformed consumers to high misfit risk. This uncertainty translates directly into costly product returns. In 2023, total merchandise returns in the U.S. retail sector reached $743 billion, with online purchases exhibiting especially high return rates—17.6% (or $247 billion) of online orders were returned (NRF, 2023). Each return is expensive: Shipping, restocking, and depreciation collectively consume 20%–65% of an item's original value (Shopify, 2025).

Despite these substantial costs, return policies remain a critical determinant of online purchase decisions. A UPS survey reports that 88% of shoppers review a retailer's return policy while shopping online, and 66% do so before completing their purchase (Brill, 2015). Return experiences strongly influence brand perception and repeat purchasing: 89% of consumers say a positive return experience increases their likelihood of making additional purchases (Appriss Retail,

[1]Antai College of Economics and Management, Shanghai Jiao Tong University, Shanghai, China
[2]Stern School of Business, New York University, New York, NY, USA

Corresponding author:
Chuanshuai Ru, Antai College of Economics and Management, Shanghai Jiao Tong University, Shanghai 200030, China.
Email: chshru@sjtu.edu.cn

2024). Recent attempts by retailers to tighten return policies have triggered consumer backlash and brand switching, as highlighted by The Wall Street Journal (Kapner, 2024). Strikingly, 87% of shoppers indicate they would likely stop buying from a brand that eliminates free returns, and 39% say they would be very likely to do so (PowerReviews, 2022). These patterns underscore a critical managerial tension: Retailers must balance consumer satisfaction and loyalty against the operational and financial burden of product returns.

Motivated by this tension, a substantial literature has examined return policy design from a theoretical perspective. Cost-recovery models, such as Su (2009), show that partial or even zero refunds maximize profit when homogeneous consumers face valuation uncertainty and sellers bear return costs. Shulman et al. (2009) similarly rationalize positive restocking fees as a tool for managing valuation uncertainty and return costs. Yet, these predictions contrast sharply with practice. Full-refund policies are pervasive across leading online retailers: Amazon offers unconditional full refunds within 30 days, Zappos provides a 365-day full-refund guarantee (Zappos, 2024), and major Chinese platforms, such as Taobao, JD.com, and Pinduoduo, routinely adopt full refund policies to build consumer trust and comply with regulatory expectations. Moreover, even identical products are sometimes listed under different combinations of prices and refund terms (Shang et al., 2017b; Taobao, 2025; Yao and Zhang, 2025), indicating that sellers use return terms as strategic contractual instruments rather than mere service policies.

This disconnect between theoretical predictions and observed practices highlights a critical gap in the literature. Most existing models overlook consumer heterogeneity arising from product misfit risk, particularly the coexistence of informed and uninformed consumers that characterizes online markets. As a result, valuation-uncertainty models alone cannot fully explain why full-refund policies persist despite high costs, nor when and why differentiated return contracts might arise. This paper seeks to bridge that gap by focusing on three key questions: Can a full-refund policy remain optimal when consumers differ in their exposure to misfit risk? How do product misfit risk and the market composition of informed and uninformed consumers jointly shape the seller's optimal return policy? Which underlying economic forces drive the emergence of contract differentiation?

To investigate these questions, we consider a monopolistic seller serving a dual-segment online market. Informed consumers know ex ante that the product fits their needs, while uninformed consumers face uncertainty about product fit at the time of purchase. The product either fits or misfits consumers. When the product fits, both consumer types share the same valuation; when it misfits, uninformed consumers derive a low valuation, simplified as zero in the base model. Because the seller cannot observe a consumer's type before offering the return policy, the natural approach is to design a menu of return contracts, each specifying a selling price and refund amount, tailored to induce self-selection. Consumers select a contract and decide whether to make a purchase. After receiving the product, they either keep or return it based on its realized valuation relative to the refund specified. Returns generate a refund obligation for the seller, whereas kept items generate the full selling price.

We first develop a parsimonious model that isolates the fundamental economic roles of return contracts. In this baseline setting, the return price mitigates misfit risk, while the selling price extracts consumer surplus. Fit products are retained and misfit products are returned. Informed consumers, who know the product fits, are indifferent to refund level and care only about selling price, whereas uninformed consumers demand generous refunds as insurance against misfit risk. We show that a single full-refund contract can implement the optimal menu because the two contract terms—the price and the refund—play complementary roles. This result is aligned with practical observations that lenient return policies are favored by both the seller and consumers, yet it contrasts sharply with prior work advocating cost-based partial refunds (Su, 2009). The full-refund policy effectively offers consumers a "free trial," allowing uninformed consumers to return misfit products at no cost. This ex-post screening enables the seller to extract the entire surplus without requiring ex ante information revelation. Although the equilibrium outcome involves a uniform contract, the menu-based framework remains essential: It verifies incentive compatibility (IC) and clarifies why such a uniform policy emerges endogenously rather than as an ad hoc assumption. Thus, even when explicit differentiation is unnecessary, the contract menu serves as the conceptual foundation that links incentive alignment, screening, and practical contract design.

When quality risk—an additional source of uncertainty in the valuation of fit products—is introduced, the analysis changes fundamentally. Full refunds become suboptimal, as consumers may receive low-quality products even when they fit. To compensate, the seller must offer a "price discount." When quality risk is low, misfit risk dominates, and the seller offers a single partial-refund contract that prevents returns of fit products by lowering both selling and return prices. In this regime, rising quality risk shifts more surplus from the seller to consumers while preserving social efficiency. However, when quality risk is high, the seller's objectives of surplus extraction and return reduction conflict. The optimal menu involves differentiated partial-refund contracts: A low-price, no-frills contract for informed consumers and a high-price, insurance-heavy contract for uninformed consumers. To sustain screening, the seller raises the return price for uninformed consumers, inducing some returns of fit products that strengthen the insurance effect. This upward distortion supports a higher selling price that deters informed consumers from imitation, consistent with the typical adverse selection result. The distortion grows with quality risk, trading off allocative efficiency for IC. Interestingly, when quality

risk becomes extreme, refunds not only mitigate information rents but also enhance social surplus by protecting consumers from retaining low-quality products. Hence, higher quality risk, though increasing return rates, can paradoxically benefit both the seller and consumers. Overall, contract differentiation arises only under high quality risk, while a simple uniform contract remains optimal in low-risk settings.

We further extend the analysis to examine other key parameters—misfit valuation, seller-side cost, and consumer return hassle cost—under misfit risk alone. The results demonstrate that the uniform contract remains robustly optimal when the fit valuation is deterministic, offering a unified explanation for a wide range of return policy practices. When the misfit valuation is low, uninformed consumers demand strong insurance, leading to a full-refund policy. When the misfit valuation is high, misfit products retain sufficient value, so refunds become less essential and the seller offers a no-refund policy, shifting full misfit risk to consumers but compensating with a lower selling price. When the misfit valuation is random, explicit contract differentiation again proves unnecessary: If informed consumers dominate the market or profit margins are thin, the seller prioritizes surplus extraction and adopts a full-refund policy; if uninformed consumers prevail or product value is high, the seller instead offers a partial refund to encourage retention of valuable misfit products, generating within-segment ex post screening. Introducing salvage value and handling costs does not alter the incentive pattern—the full-refund policy remains optimal because seller-side costs affect profits but not IC. This mechanism fundamentally differs from cost-based models such as Su (2009), where the optimal refund equals the salvage value; in our framework, the return policy manages misfit risk and consumer heterogeneity, not cost recovery. Finally, when hassle costs are low, the seller offers an overcompensating refund that covers both the selling price and hassle cost, effectively creating a "hassle-free" return experience. When hassle costs are high, they serve as a natural deterrent to returns, allowing the seller to eliminate refunds while compensating through a lower selling price.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 outlines the model framework. Section 4 characterizes the main results. Section 5 presents model extensions. Section 6 concludes with managerial implications. All proofs are provided in an e-companion.

2 Literature Review

Consumers face substantial purchase risk that arises primarily from two sources. Fit uncertainty concerns whether the product matches their needs or expectations, whereas valuation uncertainty reflects ex post variation in experienced product quality or performance. Prior research models these risks in three broad ways: (i) Bernoulli fit uncertainty, where products either fit perfectly or misfit with reduced valuation (e.g., Davis et al., 1995; Gao et al., 2023; Hsiao and Chen, 2012, 2014, 2015; Li et al., 2024; Moorthy and Srinivasan, 1995; Shulman et al., 2009); (ii) continuous valuation uncertainty without fit considerations (e.g., Akçay et al., 2013; Che, 1996; Shang et al., 2017a; Shulman et al., 2010; Su, 2009); and (iii) hybrid structures combining both sources of risk (e.g., Huang et al., 2018; Lei et al., 2022; Li et al., 2023; McWilliams, 2012; Zhang et al., 2022). Our contribution begins with a parsimonious formulation of heterogeneous fit uncertainty: Informed consumers know the product fits, whereas uninformed consumers face misfit risk. This structure isolates product misfit as the central force behind menu-based return policy design and provides a disciplined foundation for extending the analysis to environments that additionally feature valuation uncertainty.

Our study also relates to the extensive literature on optimal return policies. Early work establishes the value of money-back guarantees (full-refund policies), showing that they can stimulate demand and welfare (Che, 1996; Davis et al., 1995), signal product quality (Moorthy and Srinivasan, 1995), reduce quality differentiation (McWilliams, 2012), incentivize demand information acquisition (Taylor and Xiao, 2009), or lower marginal costs (Huang et al., 2018). More recent studies demonstrate that partial or no refunds may outperform full refunds when factoring in restocking fees (Shulman et al., 2009), strategic consumer behavior (Shang et al., 2017a), resale markets (Lei et al., 2022), or channel-level return flows (Granot and Yin, 2005; Gümüş et al., 2013; Hsiao and Chen, 2015; Su, 2009). In many of these models, including Su (2009), partial refunds arise from valuation uncertainty and seller-side costs such as salvage value or handling expenses. In contrast, our framework centers on misfit risk under consumer heterogeneity, with seller-side costs playing only a secondary role. Because return policies interact with consumers' private information, contract menus become essential. Under pure fit uncertainty, we show that a single contract—ranging from no refund to partial or full refund depending on primitives—can emerge endogenously as the profit-maximizing mechanism, even though a richer menu is feasible. When valuation uncertainty is introduced, however, differentiated partial-refund contracts may become strictly optimal. Our analysis therefore provides a foundation for the single-contract structures commonly assumed in prior work while offering a unified explanation for the full spectrum of return policies observed in online markets.

A third stream focuses on the economic roles played by return policies. Shulman et al. (2010) show that consumer return penalties can induce channel conflict: While retailers prefer high penalties to reduce return costs, manufacturers may benefit from more lenient policies that protect demand. Chen (2011) demonstrate that when consumers face both individual valuation uncertainty and aggregate demand uncertainty, optimal selling mechanisms naturally embed return features to elicit truthful consumer behaviors and resemble multi-unit auctions. Hsiao and Chen (2012, 2014) show

that refund levels can segment consumers by heterogeneous valuations or hassle costs, enabling ex post market differentiation. Several innovative practices further underscore the strategic richness of refund terms: Return-shipping insurance can signal high product quality (Zhang et al., 2022); bargaining-based refunds without returns can strictly improve performance (Gao et al., 2023); and instant return credit can boost sales while exposing sellers to fake-return risks (Li et al., 2024). Our paper contributes to this stream by establishing a screening role for return contracts that is absent in settings with only fit or valuation uncertainty. Under sufficiently high valuation uncertainty (quality risk), the seller benefits from offering a menu of return contracts that induces informed and uninformed consumers to self-select, in the spirit of classical adverse selection. Notably, under severe quality risk, refund distortions both mitigate information rents and improve social surplus by shielding consumers from retaining low-quality products.

Taken together, our study integrates and extends these streams by showing that return policies serve not only as instruments for surplus extraction and risk mitigation, but also as powerful screening mechanisms under heterogeneous fit uncertainty and valuation uncertainty. This unified perspective yields clear prescriptions on when uniform full-refund policies are optimal and when richer contract forms should arise, thereby reconciling theoretical predictions with the diversity of return practices observed in the online marketplace.

3 Model

We consider an online market where a seller sells a single product to a unit mass of consumers, consisting of two distinct segments: An informed segment and an uninformed segment. Let ρ and $1-\rho$ denote the proportions of informed and uninformed consumers, respectively. Informed consumers, due to prior positive experiences or extensive research, know with certainty the product suits them and therefore enter the market. In contrast, uninformed consumers lack prior exposure and face ex ante uncertainty about product fit. Such heterogeneity is prevalent in industries where product fit and functionality critically affect consumer satisfaction, such as apparel, housewares, and home improvements (NRF, 2022). For instance, experienced footwear buyers who are familiar with a brand or product line can reliably assess whether a shoe suits their size or comfort preferences, whereas first-time or inexperienced buyers risk misfit. To capture this risk, we assume the product either fits the consumer's needs or does not. Let $\lambda \in (0,1)$ denote the probability of a match. If the product fits, the consumer's valuation is $\tilde{\theta}$; otherwise, it is zero. Hence, informed consumers always value the product at $V_I = \tilde{\theta}$, while uninformed consumers value it at $V_N = \tilde{\theta}$ with probability λ, and $V_N = 0$ with probability $1-\lambda$. For analytical clarity, we first assume $\tilde{\theta}$ is deterministic and denote it as θ. (This restriction will be relaxed in Section 5.1.) The above setting is common knowledge. However, the seller knows only the aggregate market composition, not individual consumer types. Consumers know their own type but may have uncertain valuations of products before purchase.

The assumption that informed and uninformed consumers share the same valuation profile $\tilde{\theta}$ for a matched product reflects the premise that a product's intrinsic value—conditional on fit—is independent of informational differences. This is consistent with prior literature (e.g., Shulman et al., 2009), which assumes identical reservation values across consumer segments once the product fits. Empirical evidence also supports this view. For example, third-party organizations like Consumer Reports employ standardized methodologies to assess product performance based on objective criteria (e.g., durability, efficiency, reliability), and their findings consistently indicate that intrinsic product performance is invariant to consumers' prior knowledge or information states. Conceptually, this assumption provides a clean benchmark that isolates the informational asymmetry about product fit as the sole source of heterogeneity. It simplifies exposition without sacrificing generality; our main insights remain qualitatively unchanged when fit valuations differ across consumer types.

Given the possibility of misfit, consumers may return products for a refund if the realized valuation is low, particularly relevant for uninformed ones. To capture this, the seller announces a menu of return contracts, denoted as $\{(p_I, b_I), (p_N, b_N)\}$, tailored to the two consumer segments. Here, p_j is the per-unit selling price and $b_j(\geq 0)$ is the return price (i.e., the refund for each returned product), with $j \in \{I, N\}$ indicating informed and uninformed consumers, respectively. This framework aligns with observed practices on major e-commerce platforms. For instance, on eBay, identical products or brands may be listed with varying pricing and return policies—some offering full refunds, others none (Shang et al., 2017b). Similarly, China's national subsidy program for 3C products (Computer, Communication, and Consumer Electronics) allows consumers to choose between subsidized prices with no-return options and unsubsidized prices with return privileges on platforms such as Taobao, JD.com, and Pinduoduo (Taobao, 2025; Yao and Zhang, 2025). Abstracting from these practices, we model a profit-maximizing seller (or brand) offering two return contracts, with consumers free to choose. The seller retains the right to set partial refunds, with $b_j = 0$ or $b_j = p_j$ representing no-refund and full-refund policies, respectively. The production cost is c per unit, where $0 < c < \theta$. We assume returned products are resalable without loss of value, implying the salvage value equals c. This reflects return policies like Apple's, which allows consumers to return undamaged items in their original packaging within fourteen calendar days (Apple, 2025). Handling costs are assumed negligible (see Section 5.4 for discussion).

Faced with the contract menu, consumers either select a contract or exit the market, where the utility of the outside

option is normalized to zero. In Appendix B, we demonstrate that our main insights are robust to allowing a positive outside utility, particularly for uninformed consumers. If a consumer selects (p_j, b_j), they purchase at price p_j, then decide post-purchase whether to keep the product (deriving valuation V_j) or return it (receiving refund b_j). The expected utilities of informed and uninformed consumers under contract (p_j, b_j) are:

$$U_{Ij} = \max(V_I, b_j) - p_j = \max(\theta, b_j) - p_j,$$
$$U_{Nj} = \mathbb{E}\max(V_N, b_j) - p_j = \lambda \max(\theta, b_j) + (1-\lambda)b_j - p_j.$$

These two expressions capture the contrast between informed consumers—who know the product's fit with certainty—and uninformed consumers, who face uncertainty. The latter segment, relying on limited information about product characteristics, may return a mismatched product after purchase, as reflected by the term $(1-\lambda)b_j$. Notably, even when the product fits, a consumer may still choose to return it and derive a utility of $b_j - p_j$, provided that the return price b_j exceeds their valuation θ. This behavior is formalized by the term $\max(\theta, b_j)$. As noted by Hsiao and Chen (2012), sellers may deliberately overcompensate consumers' returns, consistent with "satisfaction guaranteed" policies observed on platforms such as Endless.com, PiperLime.com, ShoeBuy.com, and Zappos.com. Without loss of generality, we assume in cases of indifference (i.e., when the return price equals valuation), the consumer retains the product, which corresponds to the seller's preferred outcome. This assumption is also consistent with real-world behavior, where hassle costs discourage unnecessary returns. (We explicitly consider the impact of hassle costs in Section 5.5.)

The sequence of events is summarized as follows. First, the seller produces and designs a menu of return contracts. Second, consumers enter the market and choose a contract, or leave. If they leave, both parties earn zero profit or utility. Otherwise, purchases occur under the chosen contract. Upon receiving the product, uninformed consumers observe product valuation and both segments decide whether to keep or return. Then, the seller issues refunds for returned products according to the chosen contract. Finally, the seller's profit is settled.

Our model identifies product misfit risk as a core source of consumer uncertainty. While much of the literature emphasizes the importance of return policies, it often concludes that full-refund policies are suboptimal for sellers. In contrast, by incorporating consumer heterogeneity and information asymmetry, our model provides a richer treatment of uncertainty and shows that, contrary to conventional arguments, full-refund policies frequently observed in practice can indeed be optimal.

4 Analysis

This section aims to characterize the optimal menu of return contracts that maximizes the seller's expected profit. According to the revelation principle, we can, without loss of generality, focus on a truth-telling contract menu $\{(p_I, b_I), (p_N, b_N)\}$ which induces consumers to choose the contract intended for their segment $j \in \{I, N\}$.

The seller's contract design problem is formulated as:

$$\begin{aligned}
\max_{(p_I,b_I),(p_N,b_N)} \quad & \rho\left[p_I - b_I \cdot \mathbf{1}_{b_I > V_I} - c \cdot \mathbf{1}_{b_I \le V_I}\right] + (1-\rho) \\
& \left[p_N - \mathbb{E}\left[b_N \cdot \mathbf{1}_{b_N > V_N} + c \cdot \mathbf{1}_{b_N \le V_N}\right]\right] \\
\text{s.t. } & U_{II} \ge U_{IN}, && (IC_I) \\
& U_{NN} \ge U_{NI}, && (IC_N) \\
& U_{II} \ge 0, && (IR_I) \\
& U_{NN} \ge 0. && (IR_N)
\end{aligned}$$

From the objective function, each sale yields revenue p_j for the seller. The production cost c applies only when a consumer makes a purchase and retains the product (i.e., when $U_{jj} \ge 0$ and $b_j \le V_j$). Returns occur when the seller overcompensates consumers (i.e., when $b_j > V_j \ge p_j$), even if the product is a fit, that is, $V_j = \theta$. Additionally, uninformed consumers may also return the product due to misfit (with probability $1-\lambda$). The IC constraints, including (IC_I) and (IC_N), ensure that consumers prefer the truth-telling contract to the one intended for the other segment. (IR_I) and (IR_N) indicate that each consumer derives at least their reservation utility (normalized to zero), ensuring participation, that is, individual rationality (IR) constraints. If indifferent between contracts, consumers are assumed to choose their intended contract. The following proposition establishes the seller's optimal contract menu.

Proposition 1. *Among all menus of return contracts $\{(p_I, b_I), (p_N, b_N)\}$ satisfying IC and IR constraints, the following contract maximizes the seller's profit:*

$$p_I^* = p_N^* = \theta, \quad b_I^* = b_N^* = \theta.$$

Corollary 1. *The seller's optimal profit is given by $[\rho + \lambda(1-\rho)](\theta - c)$, which increases in both the proportion of informed consumers ρ and the probability of product fit λ. Both consumer segments receive zero utility.*

The analytical results in Proposition 1 deliver two fundamental insights. First, a single full-refund contract can implement the optimal menu. Second, both informed and uninformed consumers receive only their reservation utility, meaning that the seller captures the entire surplus. These results highlight complementary roles of contract terms: Mitigating misfit risk via return prices and extracting surplus via selling prices.

The optimality of the full-refund contract stems from the interaction between consumers' heterogeneous risk profiles and IC. First, the seller cannot profitably set a refund above the fit valuation θ, as this would create arbitrage opportunities that erode profit. Second, for informed consumers, who know the product fits, their utility is simply $\theta - p_I$; they never return products as long as the return price b_I is below or equal to θ. Refund terms are thus irrelevant for them, and their contract could even feature no refund at all. Since informed consumers always prefer lower selling prices, IC requires $p_N \geq p_I$ (i.e., constraint (IC_I)). Uninformed consumers, by contrast, demand insurance against misfit. Their contract must satisfy $p_N - (1-\lambda)b_N = \lambda\theta$, that is, (IR_N) binds. Any increase in p_N must be matched by a proportional increase in b_N to compensate for misfit risk. To minimize information rents, the seller sets $p_I = \theta$, forcing both p_N and b_N to equal the maximum valuation θ. The resulting full-refund contract provides complete ex post insurance to uninformed consumers. Third, both IR constraints bind, leaving both segments with zero net utility. This contrasts with standard adverse selection models, where privately informed consumers typically retain positive information rents. Here, the full-refund policy itself performs perfect screening: Uninformed consumers are effectively offered "free trials" that reveal fit without cost, eliminating the need to leave surplus. If instead uninformed consumers were offered only partial- or no-refund contracts ($b_N < p_N < \theta$), informed consumers would strictly benefit, making information rents nontrivial. This reasoning clarifies why the menu analysis is critical, even though the optimal menu ultimately collapses to a single contract. Without explicitly considering IC across multiple contracts, one could not establish why uninformed consumers require a full refund or why informed consumers are left without surplus. Thus, the menu framework—not just the resulting uniform contract—enables full surplus extraction.

Remarkably, since informed consumers never exercise returns, the seller can freely choose b_I from 0 to θ (i.e., $b_I \leq b_N = \theta$) without violating any constraint or affecting profit. Hence, a single full-refund contract (with $p_j = b_j = \theta$) is both feasible and optimal. Although the same profit could in principle be achieved by offering two contracts with identical selling prices but distinct refund terms, the full-refund contract is the only implementation that takes the form of a single contract. We emphasize this single-contract implementation for its parsimony and practical appeal: It eliminates unnecessary complexity, simplifies operations, and closely mirrors how firms design return policies in practice. Our results therefore show that explicit contract differentiation under information asymmetry may be unnecessary, while reinforcing that the menu-based approach is essential for understanding how such uniform contracts emerge. This mechanism also distinguishes return contracts from buyback contracts. Buybacks address demand risk and coordinate inventory decisions across supply chain members. In contrast, return contracts here address product misfit risk and enable surplus extraction under consumer heterogeneity. Furthermore, misfit risk in our model is segment-specific rather than valuation-driven. Unlike prior studies (e.g., Su, 2009) that advocate partial refunds to balance consumer surplus and sales volume under homogeneous valuation uncertainty, our analysis shows that when uncertainty originates from product misfit, full refunds can be optimal. This elucidates the prevalent adoption of full-refund policies in practice. Corollary 1 further characterizes the seller's profit. A fraction $\rho + \lambda(1-\rho)$ of consumers enjoy product fit and generate $\theta - c$ profit each. The remaining $(1-\rho)(1-\lambda)$, who encounter misfit, return the product and yield zero profit. As the proportion of informed consumers ρ and the probability of product fit λ increase, profit rises due to fewer returns.

These findings provide several strategic takeaways. Informed consumers care only about selling price, while uninformed consumers demand generous refunds as insurance against misfit risk. A single full-refund contract simultaneously extracts full surplus from both segments and eliminates the need for contract differentiation. Importantly, the fact that the optimal menu reduces to a uniform policy does not imply that menus are irrelevant; rather, the menu analysis is essential to verify IC and to show that uniformity arises endogenously rather than as an imposed assumption. Managers should not fixate on the specific equality $p_j = b_j = \theta$; this alignment follows from simplifying parameter choices to make the mechanism transparent. The core insight is more general: When the fit valuation is deterministic and common across consumer segments, a uniform contract structure remains robustly optimal, although the refund level may vary (see Sections 5.2–5.5). This makes the model especially relevant for functional products where misfit risk—rather than valuation heterogeneity—drives purchasing and return behavior. As we will see next, introducing randomness in the fit valuation can reinstate the need for differentiated contracts, making the contract menu once again strategically important.

5 Extensions

Proposition 1 yields two striking results: A single contract can implement the optimal menu, and this contract entails a full refund. While this outcome aligns with the widespread use of full-refund policies in practice, it rests on several simplifying assumptions—deterministic fit valuation, zero misfit valuation, fully resalable returns, zero handling costs, and no hassle costs—made to isolate the core mechanism. In reality, however, firms adopt a variety of return policies, ranging from no refund to partial to full refund. To strengthen the managerial relevance of our findings, this section systematically relaxes these baseline assumptions and explores more realistic settings. Our goal is twofold: To clarify the extent to which the uniformity and full-refund results are driven by modeling assumptions, and to provide managers with guidance on when full refunds remain optimal versus when differentiated contracts or partial refunds are desirable.

5.1 Random Fit Valuation

Our base model is well-suited to settings where only inexperienced consumers (uninformed types) face product misfit risk—for example, first-time online shoppers unable to fully evaluate product characteristics ex ante. In many contexts, however, consumers face an additional source of uncertainty: Variation in the product's performance even when the fit is correct. This can be understood as "quality risk." For example, online reviews often report that a product matches the desired style but exhibits minor defects, or alternatively that its quality is unexpectedly superior. Unlike misfit risk, quality risk affects both informed and uninformed consumers alike, as it cannot be resolved prior to purchase. Conceptually, this parallels the distinction between description uncertainty and performance uncertainty highlighted by Dimoka et al. (2012). We now examine how this additional layer of risk affects our central result—that a single full-refund contract can implement the optimal menu.

To capture the setting, we extend the base model to allow the fit valuation $\tilde{\theta}$ to be random. Specifically, let $\tilde{\theta}$ be uniformly distributed on $[\theta-\delta, \theta+\delta]$, where $\delta \in (0, \theta)$ measures the degree of quality risk. Because this risk is inherent to the product, δ is exogenous. As before, informed consumers who experience product fit have valuation $\tilde{\theta}$, whereas uninformed consumers value the product at $\tilde{\theta}$ with probability λ and zero otherwise. $\tilde{\theta}$ is revealed after the consumer receives the product. Both the seller and consumers know the distribution of $\tilde{\theta}$ ex ante (e.g., from online reviews), but not its realized value. Given a menu of contracts $\{(p_I, b_I), (p_N, b_N)\}$, the seller's expected profit is

$$\rho\left[p_I - \mathbb{E}_{\tilde{\theta}}\left[b_N \cdot \mathbf{1}_{b_N>\tilde{\theta}} + c \cdot \mathbf{1}_{b_N\le\tilde{\theta}}\right]\right] + (1-\rho)\left[p_N - \lambda\mathbb{E}_{\tilde{\theta}}\left[b_N \cdot \mathbf{1}_{b_N>\tilde{\theta}} + c \cdot \mathbf{1}_{b_N\le\tilde{\theta}}\right] - (1-\lambda)b_N\right].$$

The expected utilities of informed and uninformed consumers under contract $(p_j, b_j), j \in \{I, N\}$ are:

$$U_{Ij} = \mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b_j) - p_j,$$
$$U_{Nj} = \lambda\mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b_j) + (1-\lambda)b_j - p_j.$$

Compared with the base model, the introduction of quality risk fundamentally changes the return behavior: Both consumer segments may now return products with limited refunds even when they fit. This shift alters both the seller's profit structure and IC, suggesting that quality risk could reshape the optimal menu structure. We explore this conjecture next.

First-Best Benchmark. As a useful benchmark, consider the first-best setting where the seller can observe consumer type. The seller chooses (p_j, b_j) to maximize profit subject to consumer participation (zero net utility). Here, the selling price p_j merely transfers surplus, while the return price b_j governs efficiency. For both types, the efficient return price $b_j^o, j \in \{I, N\}$, solves

$$\max_b \quad \mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b) - \mathbb{E}_{\tilde{\theta}}\left[b \cdot \mathbf{1}_{b>\tilde{\theta}} + c \cdot \mathbf{1}_{b\le\tilde{\theta}}\right] = \mathbb{E}_{\tilde{\theta}}\left[(\tilde{\theta} - c) \cdot \mathbf{1}_{b\le\tilde{\theta}}\right],$$

which yields $b_j^o = \max(\theta - \delta, c)$. The corresponding selling prices are $p_I^o = \mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b^o)$ for informed consumers, and $p_N^o = \lambda\mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b^o) + (1-\lambda)b^o$ for uninformed consumers. Unlike in the base model, full refunds are never efficient once quality risk is present, as they induce excessive returns even from fit products. To ensure participation, the seller compensates uninformed consumers for misfit risk by offering a lower selling price; hence, $p_I^o > p_N^o$. The seller's first-best expected profit is thus:

$$[\rho + \lambda(1-\rho)]\,\mathbb{E}_{\tilde{\theta}}\left[(\tilde{\theta} - c) \cdot \mathbf{1}_{b^o\le\tilde{\theta}}\right],$$

where $\rho + \lambda(1-\rho)$ represents the fraction of consumers with product fit.

Remark 1. The seller's first-best expected profit is independent of δ when $\delta \in [0, \theta - c]$, and increases with δ when $\delta \in (\theta - c, \theta)$.

When quality risk is mild ($\delta \le \theta - c$), the optimal refund caps at the lowest possible fit valuation, $b^o \le \theta - \delta$, ensuring that no fit consumer returns the product. The seller's expected profit then remains $[\rho + \lambda(1-\rho)](\theta - c)$, with quality risk fully absorbed. When quality risk is severe ($\delta > \theta - c$), however, setting $b < c$ induces retention of low-quality products ($b < \tilde{\theta} < c$) that generate negative social surplus, thereby suppressing profit. Counterintuitively, raising the refund to $b^o = c$ is more efficient, as it screens out such loss-making products. This mechanism not only eliminates deadweight loss but also creates additional surplus, allowing the seller to benefit more as quality risk δ grows. Hence, efficiency sometimes requires deliberately inducing returns from fit but low-quality products—a sharp departure from the base model, where refunds only insure consumers but never enhance efficiency.

Second-Best Outcome (Unobservable Consumer Type). We now turn to the more realistic case in which the seller cannot observe consumer type. The seller seeks a menu $\{(p_I, b_I), (p_N, b_N)\}$ that satisfies IC and IR constraints and maximizes expected profit. Let

$$\Delta(b) = (1-\lambda)\left[\mathbb{E}_{\tilde{\theta}} \max(\tilde{\theta}, b) - b\right]$$

denote the incremental expected utility that an informed consumer derives from contract (p, b) relative to an uninformed consumer. Importantly, $\Delta(b)$ is independent of p and strictly decreasing in b for $b \le \theta + \delta$. Define the threshold values $\bar{\delta} = \frac{\lambda(1-\rho)(\theta-c)}{\lambda+2\rho-3\lambda\rho} (< \theta - c)$ and $\bar{b} = \frac{\rho(1-\lambda)(\theta+\delta)+\lambda c(1-\rho)}{\lambda+\rho-2\lambda\rho} \in (c, \theta+\delta)$. The second-best outcome can then be characterized as follows.

Proposition 2. *Suppose $\tilde{\theta}$ is uniformly distributed on $[\theta - \delta, \theta + \delta]$. Among all menus of return contracts*

$\{(p_I, b_I), (p_N, b_N)\}$ *satisfying IC and IR constraints, the following menu maximizes the seller's profit:*

(i) *If* $0 < \delta \le \bar{\delta}$, *then*

$$p_I^* = p_N^* = \theta - (1-\lambda)\delta, \quad b_I^* = b_N^* = \theta - \delta.$$

(ii) *If* $\bar{\delta} < \delta < \theta$, *then*

$$p_I^* = b^o + \frac{1}{1-\lambda}\Delta(b^o) - \Delta(\bar{b}), \quad p_N^* = \bar{b} + \frac{\lambda}{1-\lambda}\Delta(\bar{b}),$$
$$b_I^* = b^o, \quad b_N^* = \bar{b}.$$

Proposition 2 confirms our earlier conjecture: Once quality risk is introduced, a single full-refund contract is no longer optimal, and the seller cannot fully extract surplus from both consumer segments. As in standard adverse selection models, the better-informed party earns positive surplus (information rents). In our setting, informed consumers (the favorable types, fraction ρ) obtain surplus $\rho\Delta(b_N^*)$, while uninformed consumers (the unfavorable types, fraction $1-\rho$) receive only their reservation utility. The underlying intuition is as follows. In the base case without quality risk ($\delta = 0$), a full-refund contract suffices to eliminate any masquerading incentive, since $\Delta(\theta) = 0$. However, once $\delta > 0$, such a (full-refund) contract deviates from the first-best, and informed consumers are tempted to mimic uninformed ones (because $\Delta(b^o) > 0$). To prevent this, the seller must leave rents to informed consumers, making full surplus extraction infeasible.

We proceed to examine how the degree of quality risk δ shapes the optimal menu of partial-refund policies. When quality risk is low ($\delta \le \bar{\delta}$), the outcome diverges from the canonical adverse selection result. Typically, the second-best contract features upward distortion for the uninformed type but "no distortion at the top" (i.e., for the informed type). In contrast, under low quality risk, no distortion is needed at either end: $b_I^* = b_N^* = b^o$. Efficiency is preserved, and only the surplus allocation between seller and consumers shifts through the selling price. The reason is that when a fit product's performance is nearly stable, misfit risk dominates, and the seller's objectives of reducing returns and extracting surplus are aligned. One might suspect that the seller could raise b_N above $\theta - \delta$ (the lower bound of the fit-valuation distribution) to deter informed consumers from mimicking. However, doing so induces some uninformed consumers to return even fit products—an expensive outcome when δ is small, since refunds become overly generous. Instead, aligning prices ($p_I^* = p_N^*$) curbs rents more effectively without triggering unnecessary returns. Thus, the optimal menu collapses to a single partial-refund contract, as illustrated in Figure 1(a). While b_I^* can theoretically lie anywhere in $[0, \theta - \delta]$, we present the single-contract form for clarity and practical relevance. Importantly, the seller sets $b_N^* = \theta - \delta$ and charges a uniform selling price of $\theta - (1-\lambda)\delta$. This leaves each informed consumer with surplus $\Delta(\theta - \delta) = (1-\lambda)\delta$, which can be interpreted as a "quality discount": As quality risk emerges, the seller reduces refunds meant to insure uninformed consumers against misfit risk and compensates by lowering the selling price. Offering the first-best contract (p_I^o, b_I^o) to informed consumers would not help, as they would still self-select (p_N^*, b_N^*) for its lower selling price. Hence, contract differentiation again fails to yield additional profit. The extreme case $\delta = 0$ reproduces the base model's full-refund contract. As δ increases, the seller yields a surplus of $\rho\Delta(\theta - \delta)$ to informed consumers (see Figure 1(b)), and a gap of $\lambda\delta$ emerges between the selling and refund prices, implying a partial-refund policy. Nonetheless, the broader insight remains: As long as quality risk is limited, the optimal menu can still reduce to a single contract structure.

When quality risk is high ($\delta > \bar{\delta}$), contract differentiation becomes essential (see Figure 1(a)). To separate consumer types, the seller offers two distinct partial-refund contracts: A "no-frills" contract with low selling and return prices that attracts informed consumers, and an "insurance" contract with high selling and return prices that appeals to the uninformed consumers. In this regime, the typical adverse selection model result emerges: The refund for uninformed consumers (b_N^*) is distorted upward, while the refund for informed consumers remains efficient at $b_I^* = b^o$. The root cause is that with high quality risk, the twin goals of reducing returns and extracting surplus come into conflict. Recall that the efficient return price b^o suppresses unnecessary returns of fit products. When $\delta > \bar{\delta}$, however, a uniform contract would force the seller to substantially lower the selling price, granting excessive rents to informed consumers. Differentiation resolves this trade-off. By inflating b_N^* (since $\Delta(b)$ decreases with b), the seller induces more uninformed consumers to return fit but low-quality products. This strengthens the insurance effect and supports a higher selling price ($p_N^* > p_I^*$). Consequently, informed consumers, facing no misfit risk, gravitate toward the "no-frills" contract. In this way, refund distortion emerges as the core screening instrument, even at the expense of allocative efficiency. A notable implication is that return costs can actually benefit the seller. By tolerating more returns of fit products, the seller sharpens screening and extracts greater surplus. In particular, when $\bar{\delta} < \delta \le \theta - c$, the return price for informed consumers b_I^* can be set anywhere in $[0, \theta - \delta]$, since they are already discouraged from returning. Equivalently, the seller could implement two extreme contracts: A no-refund contract for informed consumers and a partial-refund contract for uninformed ones. For consistency, we fix $b_I^* = \theta - \delta$. As δ rises within this range, refund distortion intensifies, reducing both the seller's profit and informed consumers' surplus (since $\rho\Delta(\bar{b})$ falls). When quality risk becomes severe ($\delta > \theta - c$), the logic flips. Efficiency now requires setting $b_I^* = c$, even though this induces informed consumers to return. Remarkably, in this extreme case, the seller shares quality risk across both segments: Refunds to uninformed consumers mitigate information rents, while refunds to informed consumers boost social surplus.

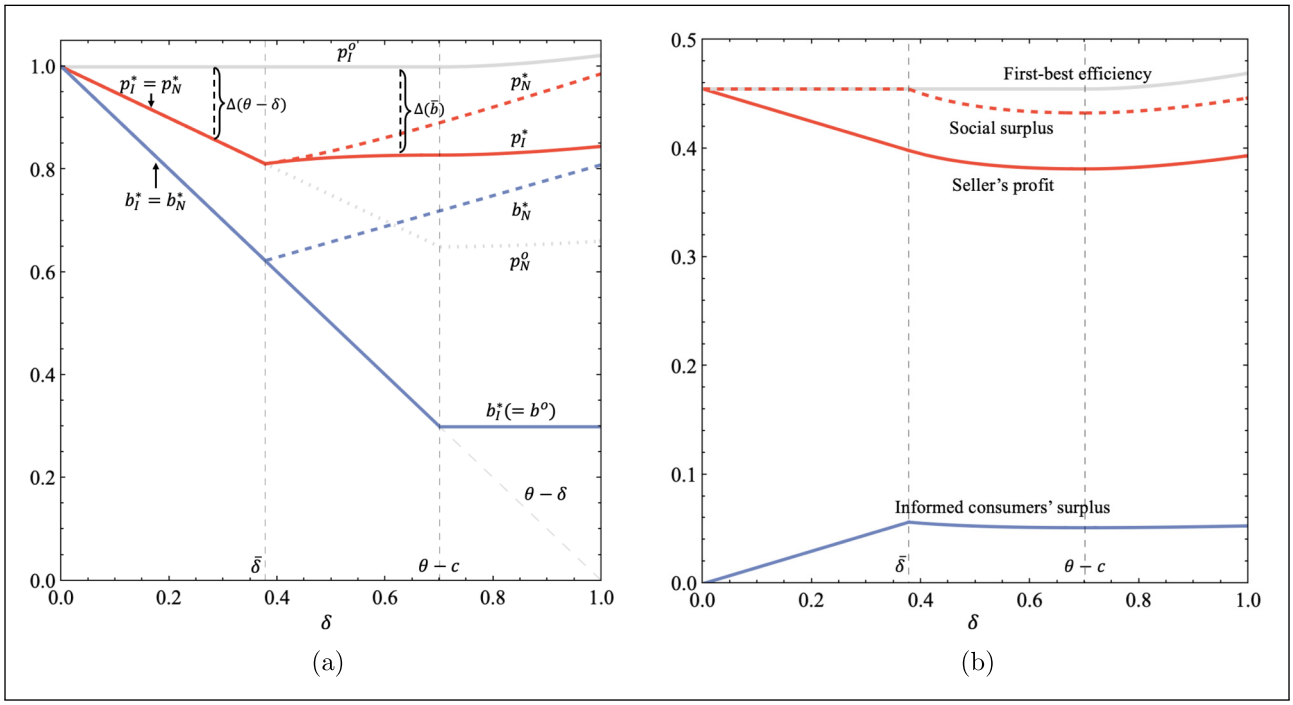


Figure 1. The second-best outcome with varying δ ($\theta = 1, \lambda = 0.5, \rho = 0.3, c = 0.3$). (a) Optimal menu of return contracts; (b) seller's profit and informed consumers' surplus.

Counter to intuition, higher quality risk here enlarges the total "pie," benefiting the seller and allowing consumers to enjoy greater surplus (Figure 1(b)).

We conclude with three primary insights from this extension. First, the base-model results prove only partially robust once quality risk is introduced. In particular, full refunds no longer appear in the optimal menu, preventing the seller from fully extracting consumer surplus. Still, a single partial-refund contract suffices when quality risk is low. Second, with high quality risk, the seller must offer differentiated return contracts: A low-price, no-frills contract for informed consumers and a high-price, insurance-heavy contract for uninformed ones. This underscores the screening role of return policies and the necessity of menus once quality risk grows substantially. Third, under severe quality risk, refunds also take on an efficiency role by protecting informed consumers against downside risk. This contrasts with the base model, where refunds were irrelevant for them. Overall, these findings enrich both the theoretical and managerial implications of our study. They explain why firms sometimes offer multiple return options for identical products, with refund generosity calibrated to consumer heterogeneity and product quality risk, while also showing that the simplicity of a single contract remains valid in low-risk settings.

5.2 Partial Misfit Valuation

In the base model, consumers were assumed to derive zero valuation from a misfit product. In reality, however, misfit items may still retain partial value. For example, a luxury fashion item purchased from an online retailer such as Farfetch might not align with a consumer's personal style yet still provide utility as a collectible, a gift, or a symbol of social status. To better capture such settings, we extend the base model by allowing uninformed consumers to derive a residual valuation θ_L from misfit products, where $0 < \theta_L < \theta$. This misfit valuation dampens their incentive to return, particularly when the refund is modest. Under a contract (p_j, b_j) for $j \in \{I, N\}$, the expected utility of an uninformed consumer becomes:

$$U_{Nj} = \lambda \max\left(\theta, b_j\right) + (1 - \lambda) \max\left(\theta_L, b_j\right) - p_j.$$

The seller's profit structure and the IC remain unchanged. Define a threshold $\hat{\theta} = \rho\theta + (1-\rho)c$. The seller's optimal contract design under partial misfit valuation is then characterized as follows.

Proposition 3. (i) *If $\theta_L \le \hat{\theta}$, the seller can offer a single full-refund contract as follows:*

$$p_I^* = p_N^* = \theta, \quad b_I^* = b_N^* = \theta.$$

(ii) *If $\hat{\theta} < \theta_L < \theta$, the seller can offer a single no-refund contract as follows:*

$$p_I^* = p_N^* = \lambda\theta + (1 - \lambda)\theta_L, \quad b_I^* = b_N^* = 0.$$

Proposition 3 shows that it remains optimal for the seller to offer a single contract when the misfit valuation θ_L is

deterministic. When the misfit product yields a low valuation ($\theta_L \leq \hat{\theta}$), the seller adopts a full-refund contract, echoing the base model's outcome. Intuitively, when misfit products deliver insufficient functionality, uninformed consumers demand strong return protection to participate. A uniform selling price equal to θ extracts all consumer surplus, leaving both segments with zero utility. However, the surplus extraction mechanism differs subtly between $\theta_L \leq c$ and $c < \theta_L \leq \hat{\theta}$. In the former case ($\theta_L \leq c$), a full-refund policy is both efficient and incentive-compatible, since all returns are socially desirable. In contrast, when $c < \theta_L \leq \hat{\theta}$, full refunds become socially inefficient because returning misfit products destroys total surplus. Under symmetric information, the seller would optimally cap the refund below θ_L to encourage product retention. Yet under asymmetric information, restricting refunds would require a lower selling price to compensate uninformed consumers for misfit risk, tempting informed consumers to mimic. To preserve IC, the seller must raise the uninformed segment's refund level, which endogenously leads to a full-refund contract as the (ex post) screening instrument.

A distinct pattern arises when the misfit valuation is high ($\theta_L > \hat{\theta}$). The seller adopts a no-refund policy for both consumer segments and lowers the uniform selling price to $\lambda\theta + (1-\lambda)\theta_L$. In this regime, informed consumers earn positive surplus (i.e., information rents), while uninformed consumers only receive their reservation utility. Here, minimizing return costs dominates the goal of minimizing information rents. Since misfit products retain sufficient value, uninformed consumers are less reliant on refund guarantees, and leveraging refund generosity for screening becomes prohibitively costly. A no-refund policy ensures all consumers retain the product, eliminating return costs and improving profit. Partial refunds with $b_N^* \leq \theta_L$ are theoretically viable in this regime but add no additional strategic benefit. The seller compensates uninformed consumers for absorbing misfit risk through a lower selling price. Because no returns occur, the refund term ceases to play a screening role, and the optimal menu collapses to a single contract that naturally grants rents to informed consumers. This mechanism is consistent with practices in markets such as Imperfect Foods and Misfits Market, where products with cosmetic imperfections but intact functionality are sold at discounted prices without refund options. These firms rely on consumers' willingness to accept minor misfit risk in exchange for lower prices—effectively implementing a low-price, no-refund policy consistent with the model's logic.

The threshold $\hat{\theta}$ increases with ρ, the fraction of informed consumers. A higher ρ strengthens the seller's incentive to eliminate information rents, favoring the full-refund regime. Conversely, when ρ is low, the seller prioritizes stimulating effective purchase from uninformed consumers by lowering the selling price and adopting a no-refund policy to avoid costly returns.

Overall, this extension reinforces the core managerial insight that a single-contract structure can remain optimal even under information asymmetry. The appropriate refund level within this contract hinges on uninformed consumers' risk tolerance, captured by the residual value of misfit products. The optimality of a full-refund policy continues to hold when the misfit valuation is low.

5.3 Random Misfit Valuation

Building on Section 5.2, we observe that when the misfit valuation θ_L is deterministic, the optimal contract menu remains uniform. Recall from Section 5.1 that introducing randomness in fit valuation can induce contract differentiation. A natural question, then, is whether uncertainty in the misfit valuation can produce similar effects. This extension demonstrates that it does not.

In many online markets, consumers are uncertain about how much value they will derive from a product that does not perfectly match their preferences, particularly for experience-based products. For instance, an online shopper may know a sweater's value if it fits but be unsure how deviations in color or texture will affect satisfaction once received. Likewise, a customer purchasing luggage online can evaluate its value when specifications suit their needs but cannot assess the durability or material quality until inspection. In such cases, the residual value of a misfit product can range from mildly disappointing to completely unsatisfactory. Unlike the uncertainty in fit valuation studied in Section 5.1—which affects both consumer segments—uncertainty in misfit valuation arises only for uninformed consumers. Informed consumers know ex ante that the product fits their needs and therefore face no misfit risk.

To capture this feature, we assume that when a misfit occurs (with probability $1-\lambda$), the uninformed consumers' residual valuation $V_N = \tilde{\theta}_L$ is uniformly distributed on $[0, \theta]$, where θ denotes the valuation under perfect fit. This distribution is common knowledge. Consistent with Section 5.1, the realized misfit valuation $\tilde{\theta}_L$ is revealed only after purchase, once the consumer experiences the product. An uninformed consumer will return the product if the realized valuation falls below the refund b_j and keep it otherwise. Accordingly, the expected utility for an uninformed consumer who accepts contract $(p_j, b_j), j \in \{I, N\}$, is:

$$U_{Nj} = \lambda \max(\theta, b_j) + (1-\lambda)\mathbb{E}_{\tilde{\theta}_L} \max(\tilde{\theta}_L, b_j) - p_j.$$

The seller's profit function and informed consumers' utility remain unaltered. With minor adjustments to (IC_N) and (IR_N), we next characterize the optimal contract menu under random misfit valuation. Define the threshold $\hat{\rho} = \frac{\theta - c}{2\theta - c}$.

Proposition 4. *Suppose the residual valuation of misfit product is uniformly distributed on* $[0, \theta]$.

(i) *If* $\hat{\rho} \le \rho < 1$*, the seller can offer a single full-refund contract as follows:*

$$p_I^* = p_N^* = \theta, \quad b_I^* = b_N^* = \theta.$$

(ii) *If* $0 < \rho < \hat{\rho}$*, the seller can offer a single partial-refund contract as follows:*

$$p_I^* = p_N^* = \frac{\theta(1+\lambda)}{2} + \frac{c^2(1-\lambda)(1-\rho)^2}{2\theta(1-2\rho)^2},$$
$$b_I^* = b_N^* = \frac{c(1-\rho)}{1-2\rho}.$$

Proposition 4 shows that contract differentiation remains unnecessary when the misfit valuation is random, yet the optimal refund level depends critically on consumer composition ρ. The underlying logic is as follows. Similar to the first-best benchmark in Section 5.1, if consumer types were observable, the seller would set the return price for uninformed consumers equal to the production cost c, ensuring that only socially inefficient (low-valuation) products are returned and fully extracting consumer surplus. However, under asymmetric information, Proposition 4 reveals that the seller must upwardly distort the return price for uninformed consumers ($b_N^* > c$) to mitigate information rents, despite the resulting efficiency loss. In contrast, the return price for informed consumers (b_I^*) remains irrelevant to profit, and thus both consumer segments face the same selling price. When informed consumers dominate the market ($\rho \ge \hat{\rho}$), the seller continues to implement a full-refund contract, paralleling the base model. In this regime, the cost of product returns is secondary because heterogeneity in misfit valuations among uninformed consumers has little impact. The seller instead prioritizes surplus extraction from informed consumers, who rarely return products, by offering a uniform selling price equal to fit valuation and maintaining IC through full refunds.

As the share of uninformed consumers rises ($1 - \rho$ increases), the seller switches to a single partial-refund contract. Relative to a full-refund policy, partial refunds allow the seller to retain additional value from uninformed consumers who keep misfit products with sufficiently high residual valuations. Although this weaker refund increases information rents and slightly lowers the selling price, the resulting reduction in costly returns more than offsets the loss. In markets dominated by uninformed consumers, limiting return costs outweighs the benefit of full surplus extraction. Importantly, unlike the no-refund scheme in Section 5.2, introducing uncertainty in misfit valuation enables the return contract to perform within-group screening among uninformed consumers: Those with low realized valuations return the product, while those with higher valuations retain it. Thus, partial refunds naturally induce cross-subsidization, where consumers who keep their purchases effectively subsidize those who return them. This mechanism extends the role of refund contracts beyond risk sharing, showing that even a single return contract embeds both incentive alignment across consumer types and screening within the uninformed group.

To further understand how market and cost parameters influence the seller's strategy, we analyze the gap between the equilibrium selling price and the return price in the following corollary.

Corollary 2. *The gap between the selling price and the return price decreases with* ρ *and* c*, and increases with* λ *(especially when* ρ *is low) and* θ*.*

As illustrated in Figure 2, the price-refund gap is largest when uninformed consumers dominate the market (low ρ). The seller sets a low return price to limit refund losses while capitalizing on consumers who retain misfit products with high residual valuations. As ρ increases, uninformed consumers become less critical, reducing the need for stringent refund control. The seller raises both the selling price and the return price to enhance surplus extraction from informed consumers and secure participation from uninformed ones, narrowing the gap. Once ρ exceeds a certain threshold, the gap disappears and the seller reverts to a full-refund policy.

The production cost c similarly narrows this gap. When c is low, the seller can sustain profitability through low selling prices and partial refunds. As c rises, tighter margins prompt the seller to increase both the selling and return prices, eventually making a full refund necessary to preserve profitability.

The product fit probability λ primarily matters when ρ is small. A higher λ boosts the expected utility of uninformed consumers, enabling the seller to raise the selling price to capture more surplus while holding the return price constant, widening the price-refund gap.

Last but not least, the fit valuation θ governs the dispersion of misfit valuations and thereby shapes the optimal refund design. When θ is low, heterogeneity in θ_L is limited, making full refunds attractive. As θ increases, the wider spread in θ_L creates opportunities for partial refunds that segment uninformed consumers by their willingness to keep misfit products, reducing returns and enhancing profitability. The model thus rationalizes an empirically observed pattern: Sellers of low-value goods (low θ) tend to offer generous refund guarantees to encourage purchase, whereas sellers of high-value products strategically impose restocking fees or partial refunds to deter frivolous returns. For example, Best Buy applies a 15% restocking fee—a form of partial refund—on high-value electronics such as drones and cameras to protect margins and reduce return frequency (Evenden, 2024).

Overall, under random misfit valuation, the optimal single-contract structure remains valid. Full refunds are preferable when informed consumers dominate or when product margins are thin; partial refunds become optimal when uninformed consumers prevail or when high-value products create opportunities for profitable within-group screening.

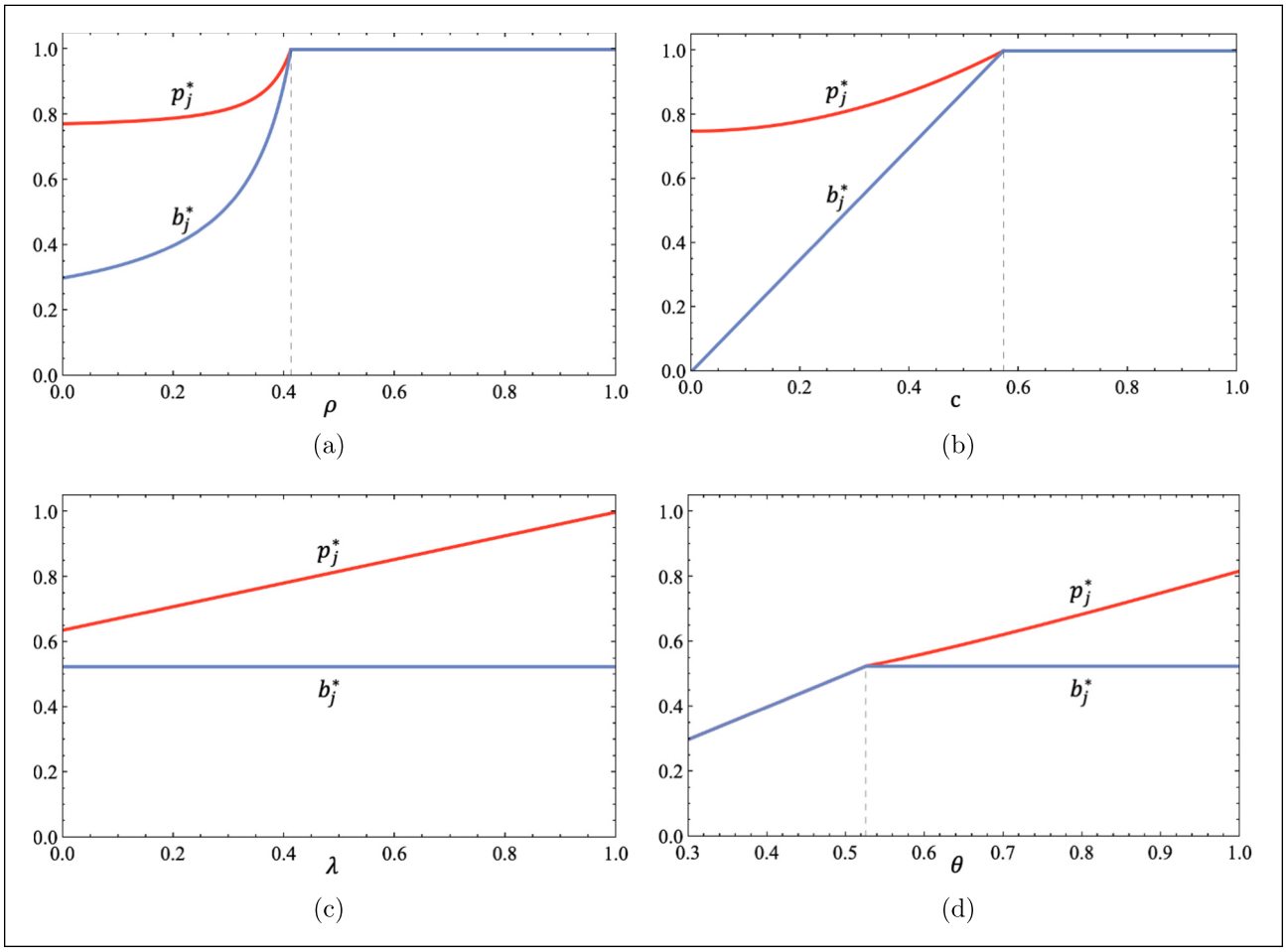


Figure 2. Optimal contract menu with varying parameters. (a) p_j^*an b_j^* with ρ ($\theta = 1, \lambda = 0.5, c = 0.3$); (b) p_j^* and b_j^* with c ($\theta = 1, \lambda = 0.5, \rho = 0.3$); (c) p_j^* and b_j^* with λ ($\theta = 1, c = 0.3, \rho = 0.3$); (d) p_j^* and b_j^* with θ ($\lambda = 0.5, c = 0.3, \rho = 0.3$).

5.4 Salvage Value and Handling Cost

In the base model, we assumed returned products are perfectly resalable without any handling cost or value loss—that is, the salvage value equals the production cost. However, in many industries, returned products must undergo inspection, refurbishment, or repackaging before resale, or they retain only partial residual value through refurbishment or material recovery. For example, Optoro, a reverse logistics technology company, estimates that processing a return typically costs firms about 66% of the product's price (Zheng, 2023). Despite this, roughly 70% of returned products remain fully functional. Apparel items, for instance, are frequently resold in bulk to wholesalers and discounters, ultimately reaching discount retailers or online outlets (Sorrel, 2025). To capture these realities, we extend the model by decoupling the production cost c from the salvage value s. In this setting, the seller's incremental cost per return changes from $b - c$ to $b - s$. The IC and consumer utility structures remain unchanged. We find that our central result continues to hold: A single full-refund contract can implement the optimal menu.

This robustness arises because the optimal return policy coordinates profits through consumer utility, which depends only on product valuation and refund terms—not on the seller's salvage value. Although the seller's profit varies with s, consumers' return decisions are driven entirely by their expected utility, which is unaffected by s. As a result, contract uniformity and full refunds persist as long as both consumer segments share the same valuation θ for a matched product. The same logic applies when returns involve a per-unit handling cost k, in which case the seller's return cost becomes $b-c+k$. In both extensions, seller-side costs alter profit levels but not the incentive structure that governs consumer screening and self-selection. This mechanism differs fundamentally from models such as Su (2009), where the optimal refund equals the salvage value (i.e., a partial refund) to prevent losses from returns. In contrast, our model demonstrates that a full-refund policy remains optimal regardless of salvage value or handling cost, because its function is not to hedge return

losses but to manage consumer fit uncertainty and information asymmetry. This distinguishes the return contract from traditional buyback or cost-based return mechanisms.

This insight helps explain the prevalence of generous return policies in practice, particularly in industries with high-salvage-value products or efficient refurbishment technologies. For example, electronics retailers often provide "no-questions-asked" returns on premium items such as smartphones and laptops. These products retain substantial resale value even when refurbished, allowing sellers to sustain profitability while fostering consumer trust and acquisition. Full refunds therefore align profitability and customer confidence when refurbishment or recycling mitigates cost exposure.

In summary, seller-side cost parameters—such as salvage value or handling cost—do not affect the optimal contract structure or necessitate contract differentiation when only misfit risk is present. Moreover, when additional uncertainty (e.g., performance risk) is incorporated, as analyzed in Sections 5.1 and 5.3, the corresponding insights remain qualitatively unchanged.

5.5 Hassle Cost

This section investigates how consumer-side return costs affect contract uniformity and refund generosity. In the base model, consumers faced no additional costs beyond the refund decision itself. In practice, however, returning products typically involves nontrivial effort. Many consumers must pay for return shipping—only 49% of online retailers offer free returns (Callarman, 2025)—or comply with platform-specific packaging and drop-off requirements. Even when retailers such as Amazon offer "no-box, no-label" return services, consumers may still bear time and transportation costs when visiting designated locations like Staples (Staff, 2024). To capture this friction, we extend the model by introducing a hassle cost $h > 0$, incurred whenever a consumer returns a product. A consumer who returns the product receives the contracted refund but pays h. Consequently, a return occurs only when the realized product valuation falls below the net return threshold $b-h$. If the refund fails to cover this cost, even mismatched products are retained. The hassle cost thus suppresses returns but may also reduce seller profitability. Define the threshold $\hat{h} = c + \frac{\rho\theta}{1-\rho}$.

Proposition 5. (i) *If $h \le \hat{h}$, the seller can offer a single contract as follows:*

$$p_I^* = p_N^* = \theta, \quad b_I^* = b_N^* = \theta + h.$$

(ii) *If $h > \hat{h}$, the seller can offer a single no-refund contract as follows:*

$$p_I^* = p_N^* = \lambda\theta, \quad b_I^* = b_N^* = 0.$$

The core insight—that a single contract implements the optimal menu—continues to hold, as informed consumers' decisions still depend only on the selling price. However, the full-refund result from the base model no longer applies once hassle costs are introduced. Surprisingly, when the hassle cost is low ($h \le \hat{h}$), the seller optimally offers an overcompensating refund, with the return price exceeding the selling price. This seemingly generous policy serves two purposes: (i) It provides full insurance against misfit risk and (ii) it reimburses consumers for the hassle cost, ensuring participation by uninformed consumers. Operationally, this mechanism resembles seller-provided return shipping insurance when h reflects return shipping expenses. Because the selling price fully extracts consumer surplus, even informed consumers gain no extra utility from this inflated refund. Crucially, the screening role of the return contract remains intact—uninformed consumers still reveal product fit ex post, without incurring any actual cost. This logic mirrors real-world practices such as Walmart's and DoorDash's doorstep return services, which eliminate packaging and shipping effort (Ryan, 2024). Empirical evidence also supports this outcome: Hassle-free returns increase repeat purchases and positive recommendations (UPS, 2013), showing that reducing return barriers can expand demand and raise profitability.

When the hassle cost becomes high ($h > \hat{h}$), the seller shifts to a no-refund policy. Covering consumers' return costs becomes too expensive, and the seller's profit motive shifts from extracting surplus to minimizing return losses. The seller thus leverages the hassle cost itself as a natural deterrent against returns and instead compensates uninformed consumers through a lower selling price. This outcome is consistent with empirical findings that high return shipping fees are a primary cause of abandoned returns (Lazar, 2024). Because no returns occur, the contract ceases to perform a screening function, forcing the seller to cede some surplus to informed consumers.

Overall, this extension reinforces one of our main managerial insights: Even with consumer-side costs, a single contract suffices to implement the optimal menu, though the refund level adapts to the magnitude of the hassle cost. In particular, when return frictions are mild, the seller optimally adopts a generous, insurance-like policy that preserves the screening role of the full-refund policy.

6 Conclusion

This paper revisits the rationale behind return policy design in online markets where a monopolistic seller faces heterogeneous consumers who differ in product misfit risk. This heterogeneity motivates the seller to offer a menu of return contracts—comprising selling and return prices—to identify consumer types. Surprisingly, the base model reveals two key findings: A single contract can implement the optimal menu, and this contract entails a full refund. This result provides a

theoretical foundation for the prevalence of full-refund policies adopted by firms such as Amazon and Zappos, where full refunds effectively act as "free trials" that reveal product fit ex post. The efficiency of this outcome arises from the complementary roles of the two contract terms: The return price insures uninformed consumers against misfit risk, while the selling price extracts surplus from informed consumers. The resulting uniformity enables the seller to sidestep the operational complexity of contract differentiation without compromising profitability.

To enhance managerial relevance, we extend the risk-profile logic to more general settings that rationalize a broader range of return policies, such as partial refunds and differentiated contracts, commonly observed in practice. This perspective distinguishes our study from prior research that attributes return policy design primarily to seller-side cost considerations (e.g., salvage value and handling cost). We show that when the product-fit valuation is deterministic, a uniform contract robustly achieves the optimal menu outcome, with refund levels shaped by the misfit valuation, market composition, and consumers' hassle cost of returns. When quality risk (i.e., randomness in product-fit valuation) is introduced, the seller prefers partial refunds. Crucially, when quality risk is high, differentiated contracts become necessary to induce consumer self-selection, revealing the screening role of return contracts. Under severe quality risk, refunds not only mitigate information rents but also enhance efficiency by protecting informed consumers against downside risk. Overall, this study offers both theoretical and managerial insights into why firms may offer multiple return options for identical products, while showing that the simplicity of a uniform contract remains optimal in low-risk environments. We conclude by discussing several limitations of our study and directions for future research. First, future research can examine richer consumer heterogeneity and more nuanced valuation structures, which may further illuminate the design of customized return policies. Second, incorporating dynamic learning and repeated purchases could help explain how consumers' information about product fit evolves over time and how return contracts adapt accordingly. Third, it would be valuable to extend the model to competitive settings, in which sellers may use return policies not only to screen consumers but also to compete for market share.

Acknowledgments

The authors sincerely thank Department Editor Albert Ha, the anonymous Senior Editor, and two anonymous reviewers for their constructive comments and suggestions, which greatly improved the paper. The authors contributed equally to this work.

ORCID iDs

Weitao Ren https://orcid.org/0009-0003-8428-0504
Chuanshuai Ru https://orcid.org/0009-0004-9993-4862
Wenqiang Xiao https://orcid.org/0000-0001-9168-3051
Fangruo Chen https://orcid.org/0000-0002-5396-5362

Funding

The authors received no financial support for the research, authorship and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online (doi: 10.1177/10591478261442318).

References

Akçay Y, Tamer B and Zhang D (2013) Selling with money-back guarantees: The impact on prices, quantities, and retail profitability. *Production and Operations Management* 22(4): 777–791.

Apple (2025) *Apple store purchase policies*. Available at: https://www.apple.com/legal/sales-support/sales-policies/retail_us.html (accessed 29 March 2026).

Appriss Retail (2024) *Returns and customer loyalty: How much do they matter?* Available at: https://apprissretail.com/blog/returns-and-customer-loyalty-how-much-do-they-matter/ (accessed 29 March 2026).

Brill J (2015) *Rethinking online returns*. Available at: https://www.ups.com/assets/resources/media/knowledge-center/UPS_ReturnsExecutiveSummary.pdf (accessed 29 March 2026).

Callarman S (2025) *A guide to ecommerce return policies in 2025*. Available at: https://www.shipbob.com/blog/ecommerce-returns (accessed 29 March 2026).

Che Y-K (1996) Customer return policies for experience goods. *The Journal of Industrial Economics* 44(1): 17–24.

Chen Y-J (2011) Optimal selling scheme for heterogeneous consumers with uncertain valuations. *Mathematics of Operations Research* 36(4): 695–720.

Davis S, Gerstner E and Hagerty M (1995) Money back guarantees in retailing: Matching products to consumer tastes. *Journal of Retailing* 71(1): 7–22.

Dimoka A, Hong Y and Pavlou PA (2012) On product uncertainty in online markets: Theory and evidence. *MIS quarterly* 36(2): 395–426.

Evenden J (2024) *Buyer beware best buy changes return policy just weeks ahead of christmas—and shoppers must watch for "restocking fees"*. Available at: https://www.the-sun.com/money/13105164/best-buy-restocking-fees-christm as (accessed 29 March 2026).

Gao Z, Huang G, Huang T, et al. (2023) A novel return policy: Negotiating refunds without returns. *SSRN*. Available at: https://ssrn.com/abstract=4834337.

Gardner M (2025) *32 Customer loyalty statistics your business needs to know in 2025*. Available at: https://emarsys.com/learn/blog/customer-loyalty-statistics (accessed 29 March 2026).

Granot D and Yin S (2005) On the effectiveness of returns policies in the price-dependent newsvendor model. *Naval Research Logistics* 52(8): 765–779.

Gümüş M, Ray S and Yin S (2013) Returns policies between channel partners for durable products. *Marketing Science* 32(4): 622–643.

Hsiao L and Chen Y-J (2012) Returns policy and quality risk in e-business. *Production and Operations Management* 21(3): 489–503.

Hsiao L and Chen Y-J (2014) Return policy: Hassle-free or your money-back guarantee? *Naval Research Logistics* 61(5): 403–417.

Hsiao L and Chen Y-J (2015) Retailer's rationale to refuse consumer returns in supply chains. *Naval Research Logistics* 62(8): 686–701.

Huang T, Ren H and Chen Y-J (2018) Consumer return policies in competitive markets: An operations perspective. *Naval Research Logistics* 65(6-7): 462–476.

Kapner S (2024) *Shoppers fight back as retailers crack down on returns*. Available at: https://www.wsj.com/business/retail/online-retail-return-policy-fees-84 44b5fe (accessed 29 March 2026).

Lazar R (2024) *7 Common return reasons and how to prevent them*. Available at: https://returngo.ai/7-common-return-reasons-and-how-to-prevent-them/ (accessed 29 March 2026).

Lei Y, Liu Q and Shum S (2022) Managing return policies with consumer resale. *Naval Research Logistics* 69(2): 241–256.

Li R, Liao C and Shi D (2024) Online retailing with instant return credit. *SSRN*. Available at: https://ssrn.com/abstract=4975151.

Li Y, Li G and Pan XA (2023) Optimal return shipping insurance policy with consumers' anticipated regret. *Production and Operations Management* 32(10): 3209–3226.

McWilliams B (2012) Money-back guarantees: Helping the low-quality retailer. *Management Science* 58(8): 1521–1524.

Moorthy S and Srinivasan K (1995) Signaling quality with a money-back guarantee: The role of transaction costs. *Marketing Science* 14(4): 442–466.

NRF (2022) *Customer returns in the retail industry 2021*. Available at: https://nrf.com/research/customer-returns-retail-industry-2021 (accessed 29 March 2026).

NRF (2023) *2023 Consumer returns in the retail industry*. Available at: https://nrf.com/research/2023-consumer-returns-retail-industry (accessed 29 March 2026).

PowerReviews (2022) *Why do shoppers return purchases, and how can brands and retailers reduce returns*. Available at: https://www.powerreviews.com/reduce-returns/ (accessed 29 March 2026).

Ryan T (2024) *Can Doorstep returns streamline the returns process?* Available at: https://retailwire.com/discussion/can-doorstep-returns-streamline-the-returns-process (accessed 29 March 2026).

Shang G, Ghosh BP and Galbreth MR (2017a) Optimal retail return policies with wardrobing. *Production and Operations Management* 26(7): 1315–1332.

Shang G, Pekgün P, Ferguson M, et al. (2017b) How much do online consumers really value free product returns? Evidence from eBay. *Journal of Operations Management* 53-56(1): 45–62.

Shopify (2025) *Ecommerce returns: Average return rate and how to reduce it*. Available at: https://www.shopify.com/enterprise/blog/ecommerce-returns (accessed 29 March 2026).

Shulman JD, Coughlan AT and Savaskan RC (2009) Optimal restocking fees and information provision in an integrated demand-supply model of product returns. *Manufacturing & Service Operations Management* 11(4): 577–594.

Shulman JD, Coughlan AT and Savaskan RC (2010) Optimal reverse channel structure for consumer product returns. *Marketing Science* 29(6): 1071–1085.

Sorrel C (2025) *What happens to your gifts after you return them?* Available at: https://www.ifixit.com/News/107532/what-happens-to-your-gifts-after-you-return-them (accessed 29 March 2026).

Staff A (2024) *Amazon offers free returns with no box, tape, or label needed*. Available at: https://www.aboutamazon.com/news/operations/free-returns-with-no-box-tape-or-label-needed (accessed 29 March 2026).

Su X (2009) Consumer returns policies and supply chain performance. *Manufacturing & Service Operations Management* 11(4): 595–612.

Taobao (2025) *Rules for the 3C product subsidy program*. Available at: https://huodong.taobao.com/wow/z/zebra/rules/dolsbkhtu1 (accessed 29 March 2026).

Taylor T and Xiao W (2009) Incentives for retailer forecasting: Rebates vs. returns. *Management Science* 55(10): 1654–1669.

UPS (2013) *UPS pulse of the online shopper: A customer experience study*. Available at: https://www.ups.com/media/en/gb/ups_global_paper.pdf (accessed 29 March 2026).

Yao K and Zhang E (2025) *China expands consumer trade-in scheme to revive economic growth*. Available at: https://www.reuters.com/world/china/china-expands-scope-consumer-trade-ins-home-appliances-2025-01-08/ (accessed 29 March 2026).

Zappos (2024) *Shipping, return & exchange policies*. Available at: https://www. zappos.com/c/shipping-and-returns (accessed 29 March 2026).

Zhang C, Yu M and Chen J (2022) Signaling quality with return insurance: Theory and empirical evidence. *Management Science* 68(8): 5847–5867.

Zheng S (2023) *The true cost of apparel returns: Alarming return rates require loss-minimization solutions*. Available at: https://coresight.com/research/the-true-cost-of-apparel-returns-alarming-return-rates-require-loss-minimization-solutions/ (accessed 29 March 2026).

How to cite this article

Ren W, Ru C, Xiao W and Chen F (2026) Why Full Refunds Prevail: A Product Fit Perspective. *Production and Operations Management* 35(10): 3715–3729.